

KARTHIKEYAN P

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience across three MERN stack internships and multiple independent and hackathon projects. Skilled in building secure, role-based, real-time web applications using React, Node.js, Express.js, and MongoDB, with working exposure to REST API design and geospatial data. Currently developing an AI-integrated hyperlocal donation platform incorporating computer vision, OCR, and route optimization. Strong collaborator with a research background in applied software engineering.

SKILLS

- **Languages:** Python, Java, C, JavaScript (ES6+)
- **Frontend:** React.js, React Native (CLI), HTML5, CSS3, Responsive Web Design
- **Backend & APIs:** Node.js, Express.js, FastAPI, REST API Design, JWT Authentication
- **Databases:** MongoDB, MongoDB Atlas (Geospatial Queries), MySQL
- **AI / Computer Vision:** YOLOv8/YOLOv11, OpenCV, SAHI, OCR, Google Gemini API, MongoDB Vector Search & RAG
- **Cloud Platforms:** Microsoft Azure Fundamentals, AWS (Security & Serverless Containers)
- **Tools & Platforms:** Git, GitHub, Gradio, ReportLab, OSRM

CERTIFICATIONS

- **Cloud:** Azure Fundamentals — CloudThat | Microsoft | NASSCOM
- **AI & Data (MongoDB):** AI & Innovation: Resilient AI Strategy; Vector Search Fundamentals (Building AI-Powered Search); Retrieval-Augmented Generation (RAG) with MongoDB; Building AI Agents with MongoDB
- **Security & Cloud (AWS):** AWS Security – Encryption Fundamentals; Choosing Serverless Containers for .NET — Infosys Springboard | AWS
- **Databases (IIT Bombay):** RDBMS PostgreSQL; PHP and MySQL — Spoken Tutorial, IIT Bombay
- **Academic (NPTEL, IIT):** Privacy & Security in Online Social Media; Cloud Computing; Ethical Hacking
- **Bootcamps & Training:** MERN Stack Certification (Vinsup Technology); Front-End Development (AWLRI, Red Rivers Labs); Generative AI Bootcamp (Raam Techlink | NoviTech R&D)

WORK EXPERIENCE

MERN Stack Development Intern

Vinsup Technology Pvt. Ltd.

Dec 2024 – Jan 2025

Madurai, India

- Built backend web modules using MongoDB, Express.js, and Node.js to support scalable, REST API-driven applications.
- Developed responsive frontend interfaces with HTML, CSS, and JavaScript, ensuring consistent behavior across devices.
- Designed, tested, and debugged REST API endpoints under industry mentor supervision, improving reliability of data exchange between frontend and backend.

Web Developer, K.L.N. Innovation & Research Park (KLN IRP)

K.L.N. College of Engineering

Dec 2024 – Present

Madurai, India

- Working full-time as the web developer on an in-house research product, owning end-to-end development within the college innovation ecosystem.
- Collaborating closely with faculty mentors and a cross-functional team to build and refine the platform toward commercial readiness.
- Designing and testing end-to-end application workflows, improving execution speed, code quality, and overall product stability.

MERN Stack Development Intern

Elysium Technologies Private Limited

Jun 2026

Madurai, India

- Completed a selective, intensive MERN Stack engineering internship, building hands-on proficiency in React.js, Node.js, Express.js, and MongoDB through applied coding exercises.
- Developed production-style full-stack components following industry coding standards and version-controlled workflows.

PROJECTS

AI Resume PDF Generator <i>Independent Lead Developer</i> - Built an end-to-end automated resume creation tool with a Gradio-based web interface, converting unstructured user input into structured resume content. - Integrated the Google Gemini API to dynamically parse and restructure raw user input into targeted, role-specific resume text. - Engineered a background PDF rendering pipeline using ReportLab, enabling instant, on-demand resume downloads.	Aug 2025
E-MBook — Digital Measurement Book System <i>Full Stack Developer (MERN) — Nimirndhu Nil Hackathon '26</i> - Developed a full-stack MERN application digitizing the manual Measurement Book process for the Rural Development & Panchayat Raj Department, replacing a paper-based workflow. - Implemented role-based authentication and a 6-tier approval workflow (JE → AE → AEE → EE → SE → CE) for hierarchical engineering approvals. - Built integrated modules for project management, measurement entry, tender allocation, and automated reporting, connecting MongoDB to Express.js REST APIs for real-time monitoring dashboards.	Jun 2026
Hospireo — Healthcare Discovery Platform <i>Full Stack Engineer</i> - Built a hospital and doctor discovery platform with secure Node.js/Express.js backend APIs supporting real-time patient booking. - Implemented MongoDB query logic for department-wise doctor profiles with real-time availability mapping. - Currently in testing phase, with deployment planned for the near term.	Testing Phase
HSDE — Hyperlocal Smart Donation Engine <i>Full Stack & AI Systems Developer (Independent / Final-Year Project)</i> - Building a mobile-centric donation platform in React Native (CLI) connecting NGOs, donors, and volunteers across 10 donation categories (food, medicines, medical equipment, blood, and more). - Architecting a custom Dynamic Donation Distribution Engine (DDSE) to match NGO requests with donors within a 15 km radius, prioritized by urgency, category, and location. - Integrating YOLO + OCR for category-aware AI verification and OSRM for optimized volunteer routing, built on FastAPI and MongoDB Atlas.	In Progress — 90%

EDUCATION

B.E., Computer Science and Engineering <i>K.L.N. College of Engineering — CGPA: 8.68 / 10 (through 6th Semester)</i>	Expected May 2027
Higher Secondary Course (HSC) <i>A.P.T. Dorairaj Higher Secondary School — 80.83%</i>	2023

ACHIEVEMENTS & LEADERSHIP

- Selected for Level 2 Ideation Camp, Nimirndhu Nil Hackathon '26 (EDII TN).
- Certified College-Level Participant, Smart India Hackathon.
- Presenter, KLNICST '26 International Conference.
- Participant, Pudhu Madurai Green MFG X Conference (CII – Madurai Zone).
- Technical Lead, Technozare Symposium.
- Technical Core Team Member, GDSC (Google Developer Student Clubs).
- Technical Team Support Member, Techgenio '26.
- Outreach Team Lead, CSE Symposium Zeigen '26.